Amitesh Puri

(+91) 7602012472 | amiteshpuri007@gmail.com | LinkedIn | GitHub | Website
West Bengal, India · 721433

Education

Generative Modelling from First Principles IISc
Ranked 6th in the final project assessment 2026

Mathematical Foundations of Machine Learning NPTEL · IISc Bangalore
Consolidated score: 68% Jan – Apr 2026

Manipal Academy of Higher Education (MIT Manipal) Udupi, Karnataka, India
M.Sc. in Applied Mathematics and Computing *CGPA: 7.28 / 10* Aug 2023 – Jun 2025

Ramnagar College, Vidyasagar University Paschim Medinipur, West Bengal, India
B.Sc. in Mathematics *Final Grade: 62%* Jul 2017 – Oct 2020

Research Experience

Physics-residual Flow Matching under distribution shift [Report] [GitHub]
Independent research report

- Compared plain flow matching, a simple weighted physics residual, and the conflict-free PBFM method on a next-step generative task.
- In distribution the weighted residual gave the best accuracy and physical consistency, lowering error by roughly a third over the baseline, while the parameter-free PBFM was no better than the baseline.
- Its advantage did not hold under stronger distribution shift, where it often reversed.

When does a geodesic probability path help latent flow matching? [Report] [GitHub]
Independent research report

- Tested whether a straight-line or a geodesic probability path changes latent flow matching, with everything else held fixed.
- Found no reliable difference between the two paths across repeated runs, and traced this to the data: path geometry helps only when the latents lie on the assumed manifold, which these did not.

Master's Research

Addressing Conditional Shift in Graph Data: An Empirical Study of Distance Metrics [Report] [GitHub]
Master's Research Report, Manipal Academy of Higher Education *Supervisor: Dr. Akansha Singh*

- Studied how semi-supervised node classification holds up under conditional distributional shift, across several GNN architectures on benchmark datasets.
- Added distance metrics as auxiliary regularisers that penalise divergence between training and test representations, and measured each metric's effect on its own.
- Used conformal prediction to check coverage under shift.

Technical Training

Guided Projects Student Nov 2022 – Feb 2023
The Machine Learning Company Remote

- Completed applied projects in NLP and computer vision, including end-to-end pipelines for text classification, sentiment analysis, and language generation.
- Trained and evaluated deep learning models for image classification and object detection tasks.

Data Science Trainee Aug 2021 – Jan 2022
Evarcity Bangalore, India

- Built Python-based data analysis and machine learning workflows on real-world datasets, with practical exposure to feature engineering, model evaluation, and data visualisation.

Personal Projects

Cyberbullying Detection [GitHub]

- Developed a text classifier in TensorFlow for identifying cyberbullying in social media content. Deployed the model through a Gradio interface, with user feedback collected and managed using Argilla for iterative data annotation.

Text Generation with GPT-2 [GitHub]

- Fine-tuned GPT-2 on a domain-specific corpus and deployed an interactive text generation interface for user evaluation and qualitative assessment of output quality.

English to Hindi Machine Translation [GitHub]

- Fine-tuned a transformer-based sequence-to-sequence model for English to Hindi translation and deployed the system through a Streamlit application with integrated text-to-speech output.

Technical Skills

Programming	Python, R
ML / DL Frameworks	PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, PyCaret
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn, Data Visualization
Tools	Git, Gradio, Streamlit, Hugging Face Hub, Argilla